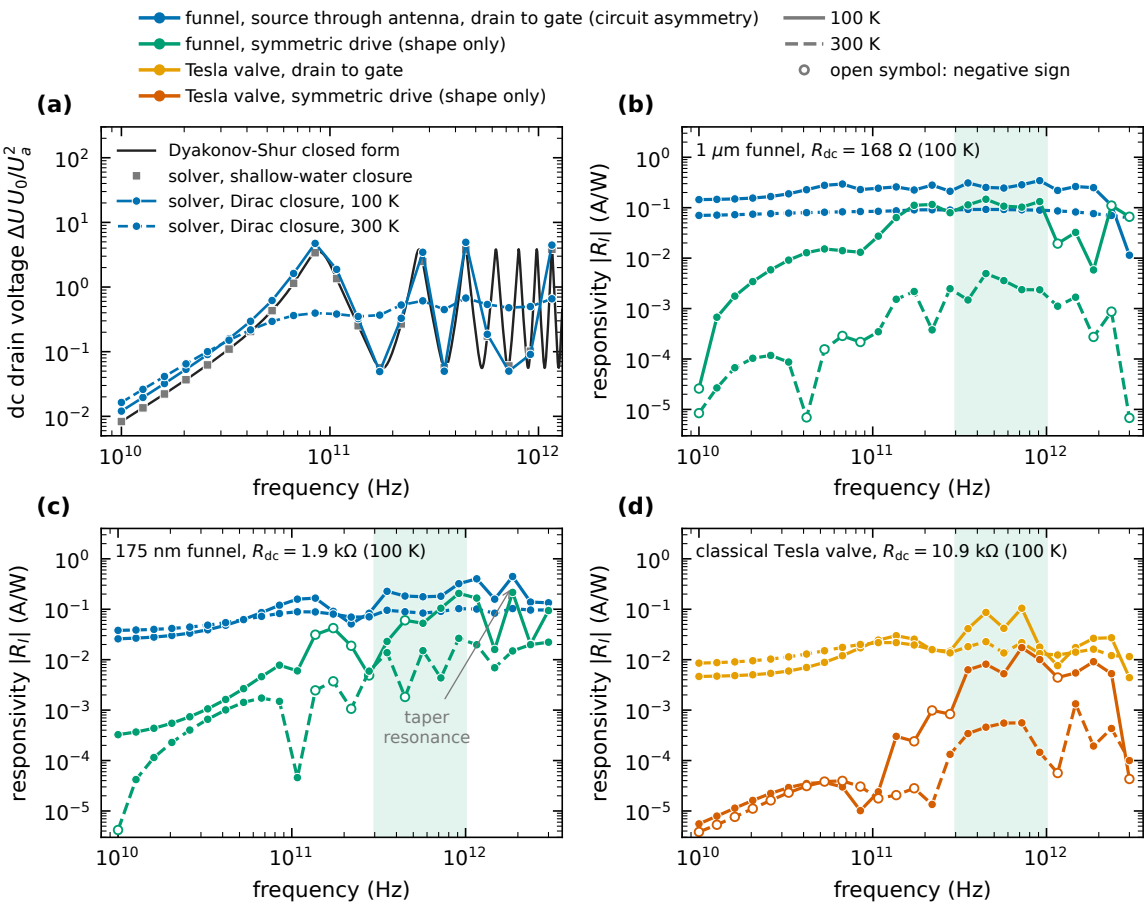
funnel, source through antenna, drain to gate (circuit asymmetry)
funnel, symmetric drive (shape only)
Tesla valve, drain to gate
Tesla valve, symmetric drive (shape only)
100 K
300 K
open symbol: negative sign
(a)
dc drain voltage $\Delta U\,U_0/U_a^2$
Dyakonov-Shur closed form
solver, shallow-water closure
solver, Dirac closure, 100 K
solver, Dirac closure, 300 K
frequency (Hz)
(b)
responsivity $|R_I|$ (A/W)
1 μm funnel, $R_{dc} = 168\ \Omega$ (100 K)
frequency (Hz)
(c)
responsivity $|R_I|$ (A/W)
175 nm funnel, $R_{dc} = 1.9$ kΩ (100 K)
taper
resonance
frequency (Hz)
(d)
responsivity $|R_I|$ (A/W)
classical Tesla valve, $R_{dc} = 10.9$ kΩ (100 K)
frequency (Hz)